Parametric, Polar and Vector-Valued Calculus — Error Analysis
Answer Key

AP Calculus BC

Quick Answers

1. $-\frac{1}{4}$
2. 12.04
3. $\frac{4\pi}{3}$
4. $-\frac{2}{3}+\frac{4\sqrt{2}}{3}$
5. 0, -1, 1
6. $\frac{2\pi}{3}$, $\frac{4\pi}{3}$
7. 1.32
8. —

Curriculum

Level: AP Calculus BC
FUN-3.A–FUN-3.C – *problems 1, 2, 5, 7, 8*
LIM-5, FUN-6 – *problems 3, 4, 6*
Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. *If they got -4: inverted the quotient and computed dx by dy instead of dy by dx*
2. *If they got 13: added the two component velocities instead of taking the magnitude of the velocity vector*
3. *If they got 25.13: integrated over the full turn from zero to two pi instead of over the single petal's bounds, sweeping the whole rose*
4. *If they got 1.667: integrated the two velocity components separately and added the results instead of integrating the speed*
5. *If they got 2.25: substituted the x-coordinate two straight into the parameter slot instead of solving for the t that reaches the point*
7. *If they got -1.87: added the two velocity components and reported the result as a speed, which came out negative*

1. Find and fix: Ana inverts the slope quotient, $x = t^2 + 2t$, $y = t^3 - 4t$, at $t = 1$.

Step 1: name the error. The chain rule gives $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$ — the parameter cancels only in that order. Ana computed the reciprocal, which is $\dfrac{dx}{dy}$, the slope of the same curve read against the wrong axis.
Step 2: differentiate each coordinate: $\dfrac{dx}{dt} = 2t + 2$ and $\dfrac{dy}{dt} = 3t^2 - 4$.
Step 3: at $t = 1$: $\dfrac{dy}{dt} = 3 - 4 = -1$ and $\dfrac{dx}{dt} = 2 + 2 = 4$, so $\dfrac{dy}{dx} = \boxed{-\frac{1}{4}}$.
Watch for: -4, the reciprocal. A slope and its reciprocal are both plausible-looking numbers, so the only defence is writing the quotient in the right order every time.

2. Find and fix: Bo's "speed" of 13 for $x = t^2 - 3t$, $y = t^3$ at $t = 2$.

Step 1: name what Bo computed. Adding x' and y' produces the sum of two components — a quantity with no geometric meaning, since the components lie along perpendicular axes. Speed is the *magnitude* of the velocity vector, which needs the Pythagorean combination.
Step 2: differentiate: $x'(t) = 2t - 3$ and $y'(t) = 3t^2$. At $t = 2$: $x'(2) = 4 - 3 = 1$ and $y'(2) = 3(4) = 12$, so the velocity vector is $\langle 1, 12 \rangle$.
Step 3: speed $= \sqrt{1^2 + 12^2} = \sqrt{145} = 12.0415\ldots$, so the speed is $\boxed{12.04}$.
Step 4: sanity check — the magnitude of a vector is at least as large as its largest component (12) and at most their sum (13). Bo's 13 is the upper bound, reachable only if the motion were along a single line.
Watch for: 13, i.e. $1 + 12$.

3. Find and fix: the area of one petal of $r = 4\cos(3\theta)$.

(a) Step 1: a petal begins and ends where the curve returns to the pole, so solve $r = 0$ first: $\cos(3\theta) = 0$ gives $3\theta = \pm\frac{\pi}{2}$, hence $\theta = -\frac{\pi}{6}$ and $\theta = \frac{\pi}{6}$. Those two consecutive zeros bound one petal.
(b) Step 2: set up the area integral over exactly that sweep:

$$A = \frac{1}{2}\int_{-\pi/6}^{\pi/6} \big(4\cos 3\theta\big)^2 d\theta = 8\int_{-\pi/6}^{\pi/6} \cos^2 3\theta \, d\theta.$$

Step 3: use $\cos^2 u = \frac{1+\cos 2u}{2}$: $8\int \frac{1+\cos 6\theta}{2} d\theta = 4\left[\theta + \frac{\sin 6\theta}{6}\right]_{-\pi/6}^{\pi/6}$. At both limits $\sin 6\theta = \sin(\pm\pi) = 0$, so only the θ term survives: $4\left(\frac{\pi}{6} + \frac{\pi}{6}\right)$.
Step 4:

$$\boxed{A = \frac{4\pi}{3} \approx 4.19}$$

Step 5: Cal's limits 0 to 2π sweep the entire rose, and for a rose with an odd petal count the full turn traces all three petals *twice* over. His 25.13 is 8π, which is six times one petal's area — not an area of any region on the page.
Watch for: 25.13.

4. Find and fix: total distance for $x = t^2$, $y = \frac{2}{3}t^3$ on $[0, 1]$.

Step 1: name the error. Distance travelled is the accumulation of *speed*, $\int_0^1 \sqrt{x'^2 + y'^2}\,dt$. Integrating the components separately and adding gives the sum of two signed displacements, which is not a length: the square root cannot be moved through a sum.
Step 2: $x'(t) = 2t$ and $y'(t) = 2t^2$, so $\sqrt{x'^2 + y'^2} = \sqrt{4t^2 + 4t^4} = 2t\sqrt{1 + t^2}$ (positive on $[0, 1]$).
Step 3: substitute $u = 1 + t^2$, $du = 2t\,dt$:

$$\int_0^1 2t\sqrt{1 + t^2}\,dt = \int_1^2 \sqrt{u}\,du = \frac{2}{3}u^{3/2}\Big|_1^2.$$

Step 4:

$$\boxed{\text{distance} = \frac{2\left(2\sqrt{2} - 1\right)}{3} \approx 1.22}$$

Step 5: why Dev's method fails in one line — his answer would be the same for a particle that reversed direction, because component integrals cancel while speed never does.
Watch for: 1.667, from $\int_0^1 2t\,dt + \int_0^1 2t^2\,dt = 1 + \frac{2}{3}$.

5. Find and fix: Elle substitutes $t = 2$ because $x = 2$, on $x = t^2 + 1$, $y = t^3 - 3t$ at $(2, -2)$.

(a) Step 1: name the error. t is a parameter, not a coordinate. The only way into the point is to solve $x(t) = 2$.
Step 2: $t^2 + 1 = 2$ gives $t^2 = 1$, so $\boxed{t = -1 \text{ or } t = 1}$. Test each against the y-coordinate: $y(1) = 1 - 3 = -2$ ✓, while $y(-1) = -1 + 3 = 2 \neq -2$. The point $(2, -2)$ is reached at $t = 1$ only.
(b) Step 3: $\dfrac{dy}{dx} = \dfrac{3t^2 - 3}{2t}$, so at $t = 1$ the numerator is $3 - 3 = 0$ and the slope is $\boxed{0}$ — the tangent there is horizontal.
Step 4: note that Elle's $t = 2$ lands on the point $(5, 2)$, a different place on the same curve entirely. Her arithmetic was flawless.
Watch for: 2.25, which is $\dfrac{3(4) - 3}{4}$ — the slope at the wrong parameter.

6. Find and fix: where does $r = 1 + 2\cos\theta$ reach the pole?

Step 1: name the error. The pole is where $r = 0$, so the equation to solve is $1 + 2\cos\theta = 0$, i.e. $\cos\theta = -\frac{1}{2}$. Ravi solved $\cos\theta = +\frac{1}{2}$, dropping the sign that the constant term forces.
Step 2: on $[0, 2\pi)$, $\cos\theta = -\frac{1}{2}$ at the two angles whose reference angle is $\frac{\pi}{3}$, in quadrants II and III.
Step 3:

$$\boxed{\theta = \frac{2\pi}{3} \quad \text{or} \quad \theta = \frac{4\pi}{3}}$$

Step 4: check: at $\theta = \frac{2\pi}{3}$, $\cos\theta = -\frac{1}{2}$ and $r = 1 + 2\left(-\frac{1}{2}\right) = 0$ ✓. These two angles are exactly the limits of the inner loop, which is why this step comes before any area integral for this curve.
Teaching point: Ravi's $\frac{\pi}{3}$ gives $r = 2$, a point far from the pole — substituting the answer back refutes it in one line.

7. Find and fix: Nia's negative "speed" for $\mathbf{r}(t) = \langle \sin 2t, \cos t \rangle$ at $t = \frac{\pi}{3}$.

Step 1: the answer is refuted before any arithmetic: speed is a magnitude, and a magnitude is never negative. Adding components can produce a negative number, which is proof the wrong operation was used.
Step 2: differentiate componentwise: $\mathbf{r}'(t) = \langle 2\cos 2t, \ -\sin t \rangle$. At $t = \frac{\pi}{3}$: $2\cos\frac{2\pi}{3} = 2\left(-\frac{1}{2}\right) = -1$ and $-\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2} \approx -0.8660$.
Step 3: speed $= |\mathbf{r}'| = \sqrt{(-1)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = \sqrt{1 + \frac{3}{4}} = \sqrt{\frac{7}{4}} = 1.3228\ldots$, so the speed is $\boxed{1.32}$.
Step 4: check — both components are negative here, so the particle is moving down and to the left, yet its speed is positive. Direction lives in the vector; speed keeps only the size.
Watch for: -1.87, the sum of the components.

8. Explain: velocity, speed, slope, and polar limits.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.
(a) *Velocity* is the vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$: it carries both how fast and in which direction the particle moves. *Speed* is the scalar $|\mathbf{v}(t)| = \sqrt{x'(t)^2 + y'(t)^2}$: the size of that vector, direction discarded, and the integrand for distance travelled. The *slope* $\dfrac{dy}{dx} = \dfrac{y'(t)}{x'(t)}$ is also a scalar, but it describes the *curve*, not the motion — it is unchanged if the particle traverses the same path twice as fast, whereas velocity and speed both double.
(b) A component of velocity is a signed rate along one axis, and the sign records direction: $x' < 0$ means leftward motion. Speed is $\sqrt{x'^2 + y'^2}$, a principal square root of a sum of squares, so it is non-negative by construction; it can only be zero, when the particle is momentarily at rest. A negative "speed" therefore always signals a wrong operation, most often adding components instead of taking the magnitude.
(c) The area formula $\frac{1}{2}\int r^2 d\theta$ accumulates the region swept by the radius. As θ runs from 0 to 2π, $r = 4\cos 3\theta$ traces all three petals of the rose, and for an odd petal count it covers each one *twice* (once with $r > 0$, once with $r < 0$, and r^2 cannot tell the difference). The full-turn integral therefore reports six petal-areas. *Habit:* before integrating, solve $r = 0$ for the θ-values that bound the region, sketch or test one interior angle to confirm the sweep covers the region once, and integrate between two consecutive zeros.
What a full-credit answer must contain: vector/scalar labelling of all three quantities, the observation that dy/dx describes the curve rather than the motion, non-negativity of a magnitude by construction, and a polar-limit habit that begins with solving $r = 0$.